Supplementary Material

StegaGuard: A Controlled Robustness Study of Lightweight Image Steganalysis under Algorithm and Payload Shift

This supplement lists the full results that the manuscript summarizes. Unless stated otherwise, values are means over seeds 42, 43 and 44 on the same 120 test source covers; they are repeated measurements on one partition, not independent samples.

Table S1. Detector × protocol × algorithm × payload results on the 120 test covers. ROC-AUC is mean ± sample SD over three seeds; F1 and BER use the validation-selected τ_{BER}; recall and FPR use the validation-selected τ_{low} (validation FPR ≤ 1%).

Detector	P	Algorithm	bpp	ROC-AUC	F1	BER	Recall@τ_{low}	FPR@τ_{low}
Residual-LR	P1	WOW	0.4	0.546 ± 0.000	0.528	0.463	0.058	0.042
			0.1	0.510 ± 0.000	0.509	0.475	0.050	0.042
		S-UNIWARD	0.4	0.565 ± 0.000	0.540	0.454	0.050	0.042
			0.1	0.505 ± 0.000	0.476	0.496	0.050	0.042
		HILL	0.4	0.570 ± 0.000	0.564	0.438	0.042	0.042
			0.1	0.514 ± 0.000	0.482	0.492	0.042	0.042
	P2	WOW	0.4	0.534 ± 0.005	0.566	0.472	0.072	0.042
			0.1	0.500 ± 0.001	0.547	0.487	0.056	0.042
		S-UNIWARD	0.4	0.549 ± 0.001	0.577	0.464	0.097	0.042
			0.1	0.504 ± 0.002	0.539	0.493	0.053	0.042
		HILL	0.4	0.551 ± 0.004	0.595	0.450	0.089	0.042
			0.1	0.505 ± 0.002	0.541	0.492	0.056	0.042
	P3	WOW	0.4	0.523 ± 0.002	0.584	0.482	0.061	0.053
			0.1	0.501 ± 0.002	0.562	0.500	0.056	0.053
		S-UNIWARD	0.4	0.555 ± 0.016	0.603	0.467	0.092	0.053
			0.1	0.502 ± 0.004	0.572	0.492	0.064	0.053
		HILL	0.4	0.546 ± 0.012	0.596	0.472	0.078	0.053
			0.1	0.511 ± 0.007	0.562	0.500	0.056	0.053
SRM+Ensemble	P1	WOW	0.4	0.714 ± 0.026	0.684	0.347	0.189	0.050
			0.1	0.539 ± 0.002	0.498	0.483	0.053	0.050
		S-UNIWARD	0.4	0.689 ± 0.008	0.669	0.360	0.114	0.050
			0.1	0.532 ± 0.015	0.501	0.482	0.061	0.050
		HILL	0.4	0.645 ± 0.009	0.636	0.386	0.108	0.050
			0.1	0.525 ± 0.009	0.501	0.482	0.056	0.050
	P2	WOW	0.4	0.713 ± 0.009	0.650	0.351	0.156	0.031
			0.1	0.550 ± 0.012	0.465	0.469	0.039	0.031
		S-UNIWARD	0.4	0.717 ± 0.006	0.648	0.353	0.097	0.031
			0.1	0.534 ± 0.009	0.464	0.471	0.025	0.031
		HILL	0.4	0.653 ± 0.018	0.585	0.394	0.072	0.031
			0.1	0.534 ± 0.007	0.459	0.475	0.028	0.031
	P3	WOW	0.4	0.686 ± 0.028	0.647	0.358	0.147	0.022
			0.1	0.546 ± 0.016	0.474	0.469	0.033	0.022
		S-UNIWARD	0.4	0.677 ± 0.022	0.610	0.383	0.131	0.022
			0.1	0.528 ± 0.005	0.450	0.485	0.019	0.022
		HILL	0.4	0.632 ± 0.010	0.586	0.400	0.083	0.022
			0.1	0.517 ± 0.010	0.445	0.489	0.036	0.022

Residual-LR shows a seed SD of 0.000 under P1, consistent with P1 involving no variant sampling and a deterministic fit. The FPR column is constant within each detector and protocol because the 120 test covers are scored identically in all six stego conditions. Table S3 gives the per-run false-positive counts behind these rates.

Table S2. Payload-shift comparisons (RQ3). ΔAUC = AUC(0.1 bpp) − AUC(0.4 bpp), paired on the 120 test covers, with 95% paired bootstrap intervals (resampling unit: source cover; one draw shared by the three seeds; B = 10,000; see Sect. 3.5 of the manuscript). ΔAUC is computed from unrounded AUCs and may differ by 0.001 from the rounded columns.

Detector	P	Algorithm	AUC 0.4	AUC 0.1	ΔAUC	95% CI
Residual-LR	P1	WOW	0.546	0.510	−0.036	[−0.064, −0.007]
		S-UNIWARD	0.565	0.505	−0.060	[−0.093, −0.029]
		HILL	0.570	0.514	−0.056	[−0.095, −0.021]
	P2	WOW	0.534	0.500	−0.034	[−0.058, −0.011]
		S-UNIWARD	0.549	0.504	−0.045	[−0.070, −0.022]
		HILL	0.551	0.505	−0.046	[−0.074, −0.020]
	P3	WOW	0.523	0.501	−0.022	[−0.046, 0.000]
		S-UNIWARD	0.555	0.502	−0.054	[−0.079, −0.031]
		HILL	0.546	0.511	−0.035	[−0.065, −0.006]
SRM+Ensemble	P1	WOW	0.714	0.539	−0.175	[−0.207, −0.143]
		S-UNIWARD	0.689	0.532	−0.157	[−0.188, −0.127]
		HILL	0.645	0.525	−0.120	[−0.149, −0.091]
	P2	WOW	0.713	0.550	−0.163	[−0.193, −0.134]
		S-UNIWARD	0.717	0.534	−0.183	[−0.212, −0.154]
		HILL	0.653	0.534	−0.119	[−0.149, −0.088]
	P3	WOW	0.686	0.546	−0.140	[−0.170, −0.111]
		S-UNIWARD	0.677	0.528	−0.149	[−0.178, −0.120]
		HILL	0.632	0.517	−0.115	[−0.144, −0.087]

Table S3. Test false positives at τ_{low} in each run (of 120 test covers), counted from the per-image test scores. The test FPR is the total divided by the 3 × 120 = 360 cover evaluations.

Detector	P	Seed 42	Seed 43	Seed 44	Total	Test FPR
Residual-LR	P1	5	5	5	15/360	0.042
	P2	4	6	5	15/360	0.042
	P3	6	7	6	19/360	0.053
SRM+Ensemble	P1	7	3	8	18/360	0.050
	P2	7	2	2	11/360	0.031
	P3	5	1	2	8/360	0.022

Software. All experimental results in the manuscript and in Tables S1-S3 were computed from the 600 source images with Python 3.13, NumPy 2.5.3 (OpenBLAS, Haswell kernel), SciPy 1.18.1, scikit-learn 1.9.1, Pillow 12.3.0, conseal 2025.11 and sealwatch 2025.9.